05 The Selection Room

Wright–Fisher + selection

CORE PATH · BSC	EXTENSION · MSC	BEFORE THIS ROOM
about 35 min	**+ 32 min**	The Drift Room (04): this room is the same sampling with one allele favoured.
steps 1–4	steps 5–7	

OPEN THE ROOM

`https://evolinus.github.io/zooe/teaching/evolutionary-laboratories/#selection`

Allele A1 now carries a fitness advantage *s*. Each generation the frequency is reweighted by fitness, and then sampled at random exactly as in the Drift Room. This guide asks when selection wins that contest, and when chance does.

LEARNING OBJECTIVES

After the core path you should be able to

- Explain how a selective advantage changes the expected allele frequency each generation, and why the realised path still wanders.
- Predict how population size changes how reliably a favoured allele is fixed.
- Compare dominant, additive and recessive advantages in a diploid population.
- State the rule of thumb that the product *Ns*, not *s* alone, decides whether selection or drift dominates.

The extension adds

- Use Kimura's *u(p)* to predict fixation probabilities and test them against batches of runs.
- Explain, using Hardy–Weinberg proportions, why a rare recessive allele is nearly invisible to selection.
- Use $E[T_{fix}] \approx (2/s) \ln M$ to predict how long a sweep takes.

SUGGESTED ROUTES

- **45-minute practical (BSc):** steps 1–4 and questions 1–3.
- **Two-room session (90 minutes):** the Drift Room core path, then this one.
- **MSc:** steps 5–7 and questions 4–6. Step 6 pairs well with the Hardy–Weinberg Room.

How to work through it. Every step asks you to *predict* before you press anything, and to *explain* afterwards. Write the prediction down and commit to it: a wrong prediction you can account for teaches more than a right one you guessed. Runs are random, so where a step says so, run it more than once before you decide what happened.

1 Selection in a large population

10 min

SET UP
Haploid, N = **200**, p = 0.50, s = **0.10**.

PREDICT FIRST
The dashed line shows what selection alone would do. Will the real run follow it closely or wander far from it? About how many generations until A1 is nearly fixed?

RUN
Press **Run Simulation**. Then run again.

LOOK AT
the solid trajectory against the dashed line, and the *Δp (selection shift)* readout.

EXPLAIN
Why does a large population track the deterministic line so closely?

2 The same advantage in a small population

10 min

SET UP
Haploid, N = **10**, p = 0.50, s = 0.10.

PREDICT FIRST
A1 is still 10% better. Out of ten runs, how many will end with A1 fixed?

RUN
Press **Run 10 Simulations**, twice, and count the runs in which A1 was lost.

LOOK AT
how far the ten trajectories stray from the dashed line.

EXPLAIN
How can a genuinely better allele be lost? Use the words *sampling* and *population size* in your answer.

3 Dominance

10 min

SET UP

Diploid, $N = 50$, $p = 0.50$, $s = \mathbf{0.20}$. Compare $h = \mathbf{1}$ (A1 dominant) with $h = \mathbf{0}$ (A1 recessive).

PREDICT FIRST

Sketch both trajectories. Which rises faster at the start? Which takes longer to finish the job?

RUN

Run each setting twice, and compare the shapes of the curves.

EXPLAIN

When A1 is common, most A2 copies sit in heterozygotes. Use that to explain why the dominant allele slows down near fixation — and why the recessive one is slow at the start.

4 Switch selection off

5 min

SET UP

$s = \mathbf{0}$.

PREDICT FIRST

Which room is this now, and what will the dashed line do?

RUN

Run once.

EXPLAIN

Why is it useful to have the neutral case available inside the same room?

EXTENSION · MSC

5 Kimura's prediction

12 min

SET UP

Haploid, $s = 0.10$, starting from a single copy: $N = 10$ with $p = 0.10$, then $N = 100$ with $p = 0.01$.

PREDICT FIRST

Use $u(p) = (1 - e^{-2Nsp})/(1 - e^{-2Ns})$ to calculate the fixation probability in each case, and the neutral value p for comparison. By what factor does the advantage multiply the chance of fixation in each population?

RUN

Press **Run 10 Simulations** three times at each setting and count fixations.

EXPLAIN

Your counts are small samples; are they consistent with your predictions? Why does the same s multiply the chance by so much more in the larger population?

6 A rare recessive advantage

12 min

SET UP

Diploid, $N = 100$, $p = \mathbf{0.05}$, $s = 0.20$. Run with $h = 1$, then $h = 0.5$, then $h = 0$.

PREDICT FIRST

Rank the three in how often A1 will be fixed. Roughly what fraction of runs will fix the recessive allele?

RUN

Run 10 Simulations twice at each h and record fixations out of 20.

EXPLAIN

At $p = 0.05$, what fraction of A1 copies sit in heterozygotes? Why does that make h matter so much while the allele is rare?

7 How long does a sweep take?

8 min

SET UP

Haploid, $N = 200$, $p = 0.01$; $s = 0.05$, then $s = 0.20$.

PREDICT FIRST

Use $\mathrm{E}[T_{\mathrm{fix}}] \approx (2/s) \ln M$ to predict the time a successful sweep takes at each s.

RUN

Run single simulations until you have one fixation at each s; note the generation it fixed.

EXPLAIN

Why does the time depend on $1/s$ but only on the logarithm of M?

DISCUSSION AND WORKSHEET QUESTIONS

1. BSc A new antibiotic-resistance mutation gives a 5% advantage. Is it more likely to spread in a population of 10^9 bacteria or in a population of 50 birds? Explain.
2. BSc "Natural selection always picks the best allele." Give two reasons from this room why that is not quite true.
3. BSc Why does the room say the N in Ns should really be an effective population size?
4. MSc Derive why $u \approx 2s$ for a single new beneficial copy when $Ns \gg 1$, and why $u \approx 1/N$ when $|Ns| \ll 1$.
5. MSc Many human disease alleles are recessive and rare. Explain why selection against them is so slow, using step 6.
6. MSc How would you expect a slightly *harmful* allele ($s = -0.01$) to behave at $N = 20$? At $N = 2000$?

CHECKING YOUR REASONING

Read this after you have written your own answers. These are what the model should show on most runs — not the only acceptable explanations.

- **Step 1.** At $N = 200$ the run hugs the dashed line; A_1 is close to fixation after about 50 generations (its odds grow by 10% a generation, and $\ln 99 / \ln 1.1 \approx 48$).
- **Step 2.** At $N = 10$, A_1 is lost in roughly a quarter of runs from $p = 0.5$ (Kimura's formula gives fixation 0.73). Sampling noise is as large as the selective push.
- **Step 3.** $h = 1$ rises fast early but crawls at the end, because the last A_2 copies hide in heterozygotes where they cost nothing. $h = 0$ starts slowly and finishes fast.
- **Step 4.** With $s = 0$ the room is the Drift Room; the dashed line is flat at the starting p.
- **Step 5.** $N = 10$, $p = 0.1$: $u \approx 0.21$ against 0.10 neutral — the advantage only doubles the chance. $N = 100$, $p = 0.01$: $u \approx 0.18$ against 0.01 neutral — eighteen times. In a larger population drift is weaker, so the same s counts for more; three batches of ten (30 runs) will match only roughly.
- **Step 6.** In 20 runs expect nearly all to fix at $h = 1$, most at $h = 0.5$, and only about a third at $h = 0$. At $p = 0.05$, 95% of A_1 copies are in heterozygotes, where a recessive advantage is invisible.
- **Step 7.** With $M = 200$: about 210 generations at $s = 0.05$ and about 53 at $s = 0.2$, give or take a lot for single runs.